

Collecting and Storing Data

Sources, ingestion patterns, storage services, formats, and scalability for ML and AI



Domain 1 begins with reliable data collection

Why collection matters

Models need relevant, permitted, and accessible data. Weak collection decisions can create missing records, delays, higher cost, or security problems.

What MLA-C02 expects

Choose sources, ingestion patterns, storage services, formats, and scaling methods based on the data and business requirements.



Batch and streaming solve different timing needs

Batch ingestion

Data moves on a schedule or in large groups. It suits historical files, regular reports, and workloads where a delay is acceptable.

Streaming ingestion

Events arrive continuously and require quick processing. Amazon Kinesis, Amazon MSK, and Apache Flink support near real-time pipelines.



Storage choices follow the access pattern

01 Amazon S3

Object storage for data lakes, files, datasets, model artifacts, and multimodal content.

02 Amazon EBS or EFS

Block or shared file access when compute needs disk-like storage or a shared file system.

03 RDS or DynamoDB

Managed databases for relational queries or low-latency key-value access.

04 Search and vectors

OpenSearch and other vector stores support text search, embeddings, and AI retrieval.



File formats change performance and cost

	CSV	JSON	Parquet / ORC
Structure	Tabular	Nested or flexible	Columnar
Best fit	Simple exchange	Events and APIs	Analytics and ML
Main tradeoff	Large and loosely typed	Readable but verbose	Efficient but less human-readable
Access pattern	Read complete rows	Read nested records	Read selected columns



AI systems collect more than tables

01 Structured data

Rows and columns from databases, transactions, customer systems, and business applications.

02 Text

Documents, messages, support records, prompts, responses, and metadata.

03 Images and audio

Media files require storage, labels, permissions, and suitable preprocessing.

04 Embeddings

Numerical vectors represent meaning and support similarity search in RAG and other AI systems.



A dependable ingestion path has five checkpoints

01

02

03

04

05

Source

Confirm availability, ownership, schema, and permissions.

Transfer

Match batch or streaming movement to volume and freshness.

Validate

Check record counts, formats, errors, and late data.

Store

Partition and organize data for later access.

Observe

Monitor capacity, failures, lag, throughput, and cost.



Collection creates the foundation for every later decision

**Match ingestion to data arrival and freshness needs.
Choose storage by access pattern, scale, and
compliance.**

**Select formats that support efficient downstream
use.**

Monitor the full path from source to destination.

NEXT

**Part 2:
Transformation and
feature engineering**

